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# Reconfigurable Fault-tolerant NMPC for Y6 Coaxial Tricopter with Complete Loss of One Rotor

Mohit Mehndiratta and Erdal Kayacan

*Abstract*— **Safe and reliable operation of unmanned aerial vehicles (UAVs) is a high priority as most of the UAVs must be operated in humanly populated environments. In this study, we exhibit a nonlinear model predictive control (NMPC) approach, as an active fault-tolerant control method, for a Y6 coaxial tricopter UAV in the presence of an actuator fault respecting the actuator constraints. A cascaded closed-loop control methodology, incorporating a reconfigurable low-level controller, is proposed which eradicates the need to design multiple controllers for normal and faulty operations. The efficacy of the presented methodology is illustrated for tracking an 8-shaped trajectory with a complete loss of one rotor. From the sequential failure cases of two different rotors, it is shown that even with a fault-detection delay of $0.5$s, the maximum altitude divergence is $0.41$m that smoothly diminishes over the span of few seconds. In addition to the adequate tracking performance, thanks to ACADO toolkit, a combined average execution time of $2.1$ milliseconds is achieved which is promising for its real-time applicability for fast aerospace applications.**

## I. Introduction

Despite various impressive real-world applications of multirotor unmanned aerial vehicles (UAVs), including but not limited to, medical product transportation [1], search and rescue at disaster sites [2], and tracking of moving ground targets [3]; their safe and reliable operation in a humanly populated environment has always been a major concern. Additionally, this concern is regarded as a bottleneck to their comprehensive integration into civil airspace. Nevertheless, in order to ensure their safe operation in populated environments, many fault-tolerant control (FTC) strategies have been successfully implemented including, control of quadrotor while losing one, two and three rotors in [4], control of an octorotor with a rotor failure in [5], and trajectory replanning for fault-tolerant formation control of quadrotor UAV in [6]. In general, two types of FTC exist: passive and active [7]. While the controllers are fixed and designed to be robust against a certain class of presumed faults in the former, the latter employs controllers with reconfiguration ability, incorporating the fault information [8]. Moreover, on account of limited fault handling capability of the passive approach, the active FTC method is favored in this study.

Model predictive control (MPC) has been identified as a promising candidate for reconfigurable (or active) FTC [9], [10]. This is on account of its ability to systematically handle most of the challenges of reconfiguration, namely: (i) accommodation for changing process dynamics via recursive online optimization, (ii) operation with system constraints, and (iii) applicability to nonlinear systems in terms of nonlinear MPC (NMPC). Recent applications of MPC include learning-based control of ground and aerial robots in [11]–[13], control of a quadrotor for aerial pick-and-place tasks in [14], and control of a tilt-rotor tricopter in [15]. Meanwhile, MPC has been applied for FTC of numerous systems including a wind turbine benchmark model with various actuators and component-wise faults in [16], an F-16 aircraft model with control surface faults in [17], and a quadrotor UAV with actuator faults in [18]–[20]. Another active FTC application of MPC based on fuzzy logic has been demonstrated for the control of wind turbine with actuator faults in [21]. Amongst the works in [19] and [20], MPC has been utilized for FTC of quadrotor UAV with reduced actuator effectiveness, i.e., with partial actuator failures. Although the case with a failure of one motor is studied in [18], a successful yaw, pitch and roll control could not be achieved simultaneously due to insufficient redundancy of the actuators in the quadrotor.

In this paper, we demonstrate a cascaded NMPC structure for the FTC application of a Y6 coaxial tricopter UAV with a complete loss of one rotor. The cascaded structure comprises of a position controller at high-level and an attitude controller at low-level. The importance of utilizing a cascaded NMPC structure over a full-state controller can be acknowledged in terms of: (i) ability to distribute the overall control task such that the two NMPCs can run on different low-cost embedded hardware, instead of running one but on an expensive onboard computer, and (ii) flexibility to run the two loops at different frequencies. The latter aspect is important for UAV control as the low-level controller providing stability has to run at a comparatively higher rate. In summary, the contributions of this paper are as follows:

- Design of a cascaded NMPC framework for the FTC of a tricopter UAV with a complete loss of one rotor.
- Reconfigurable low-level NMPC which eliminates the need to employ multiple controllers for normal and faulty operations.
- Elaboration of the tracking performances for various fault detection delay times.

In order to focus on the reconfigurable fault-tolerant NMPC in this paper, no explicit fault-detection method is implemented. However, for real-time applicability of the closed-loop, the low-level controller is designed to tackle the fault-detection delay time upto $0.5$s, which is considered to be reasonable for fast UAV applications [5].

This paper is organized as follows: Section II starts with

Mohit Mehndiratta is with the School of Mechanical and Aerospace Engineering, Nanyang Technological University, 639798, Singapore. Email: mohit005@e.ntu.edu.sg

Erdal Kayacan is with the Department of Engineering, Aarhus University, Aarhus 8200, Denmark. Email: erdal@eng.au.dk

dynamic modeling of the Y6 coaxial tricopter. In Section III, NMPC problem formulation is presented along with the design of each component of the closed-loop framework. Numerical results are illustrated in Section IV. Lastly, in Section V, some conclusions are drawn from this study.

## II. Y6 Coaxial Tricopter Model

The coaxial tricopter consisting of three arms, each having two coaxial rotors, is considered. This configuration is favored over a quadrotor UAV due to its actuator redundancy. Its free-body diagram is shown in Fig. 1 along with the assigned coordinate frames. In the considered configuration, the top three rotor – rotor 1, rotor 3 and rotor 5 – are rotating clockwise, while the bottom three rotors – rotor 2, rotor 4 and rotor 6 – are rotating counter-clockwise.

### A. Kinematic Equations

The kinematic equations, obtained by using the transformation from the body frame ($\mathcal{F}_B$) to the Earth-fixed frame ($\mathcal{F}_E$), can be written as:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} = R_{EB} \begin{bmatrix} u \\ v \\ w \end{bmatrix}, \quad \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = T_{EB} \begin{bmatrix} p \\ q \\ r \end{bmatrix}, \quad (1)$$

where $x$, $y$, $z$ and $\phi$, $\theta$, $\psi$ are the translational position and rotational attitude, respectively, that are defined in frame $\mathcal{F}_E$; $u$, $v$, $w$ and $p$, $q$, $r$ are the translational and rotational velocities that are defined in frame $\mathcal{F}_B$; $R_{EB}$ is the translation transformation matrix between frames $\mathcal{F}_E$ and $\mathcal{F}_B$, while $T_{EB}$ maps the rotational velocity component from $\mathcal{F}_B$ to $\mathcal{F}_E$ ($c : cos$, $s : sin$, $t : tan$):

$$R_{EB} = \begin{bmatrix} c\theta c\psi & s\phi s\theta c\psi - s\psi c\phi & c\phi s\theta c\psi + s\phi s\psi \\ c\theta s\psi & s\phi s\theta s\psi + c\psi c\phi & c\phi s\theta s\psi - s\phi c\psi \\ -s\theta & s\phi c\theta & c\phi c\theta \end{bmatrix}, \quad (2)$$

$$T_{EB} = \begin{bmatrix} 1 & s\phi t\theta & c\phi t\theta \\ 0 & c\phi & -s\phi \\ 0 & \frac{s\phi}{c\theta} & \frac{c\phi}{c\theta} \end{bmatrix}. \quad (3)$$

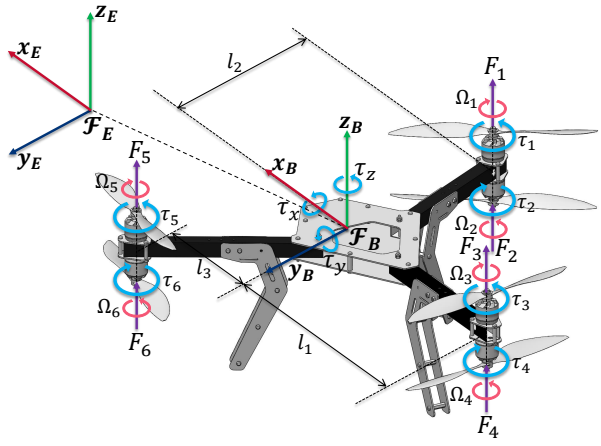



Fig. 1: Coordinate frame and sign conventions for the considered Y6 coaxial tricopter UAV.

### B. Rigid-body Equations

The rigid-body dynamic equations are derived based on Newton-Euler formulation in $\mathcal{F}_B$ frame, along with an assumption that the point-mass is present at the center of gravity [22]. They are categorized as

*Force Equations:*

$$\dot{u} = rv - qw + gsin(\theta) + \frac{1}{m}F_x, \quad (4a)$$

$$\dot{v} = pw - ru - gsin(\phi)cos(\theta) + \frac{1}{m}F_y, \quad (4b)$$

$$\dot{w} = qu - pv - gcos(\phi)cos(\theta) + \frac{1}{m}F_z, \quad (4c)$$

*Moment Equations:*

$$\dot{p} = \Big(\frac{1}{I_{xx}I_{zz} - I_{xz}^2}\Big)\Big[\{-pq(I_{xz}) + qr(I_{yy} - I_{zz})\}I_{zz} - \{qr(I_{xz}) + pq(I_{xx} - I_{yy})\}I_{xz} + \tau_x(I_{zz}) - \tau_z(I_{xz})\Big], \quad (5a)$$

$$\dot{q} = pr\Big(\frac{I_{zz} - I_{xx}}{I_{yy}}\Big) - (r^2 - p^2)\Big(\frac{I_{xz}}{I_{yy}}\Big) + \tau_y\Big(\frac{1}{I_{yy}}\Big), \quad (5b)$$

$$\dot{r} = \Big(\frac{1}{I_{xx}I_{zz} - I_{xz}^2}\Big)\Big[\{qr(I_{xz}) + pq(I_{xx} - I_{yy})\}I_{xx} - \{-pq(I_{xz}) + qr(I_{yy} - I_{zz})\}I_{xz} + \tau_z(I_{xx}) - \tau_x(I_{xz})\Big], \quad (5c)$$

where $F_x$, $F_y$, $F_z$ are the total external forces and $\tau_x$, $\tau_y$, $\tau_z$ are the total external moments. In addition, $I_{xx}$, $I_{yy}$, $I_{zz}$, and $I_{xz}$ represent the moments of inertia of the tricopter along axes $\mathcal{F}_{B_x}$, $\mathcal{F}_{B_y}$, $\mathcal{F}_{B_z}$, and $\mathcal{F}_{B_{xz}}$, respectively. It is to be noted that unlike a quadrotor UAV, the coaxial tricopter only has a single plane of symmetry, and hence, the effect of asymmetric moment $I_{xz}$ is explicitly considered in (5).

The external aerodynamic forces and moments, in terms of steady-state thrust and drag-moment, generated by a hovering rotor can be modeled utilizing the momentum theory as:

$$F_i = K_F\Omega_i^2 \quad \text{and} \quad \tau_i = K_\tau\Omega_i^2, \quad (6)$$

where $F_i$ and $\tau_i$ are the force and drag-moment generated by the $i^{\text{th}}$ rotor, and $K_F$ and $K_\tau$ are some positive intrinsic parameters that are commonly referred as the force and drag-moment coefficients, respectively. Also, the effective angular velocity $\Omega_i$ is modeled as:

$$\Omega_i = \alpha_i\omega_i, \quad (7)$$

where $\omega_i$ represents the rotational velocity of the $i^{\text{th}}$ rotor and the parameter $\alpha_i \in [0, 1]$ captures the fault information, wherein its 1 and 0 values represent normal and faulty conditions, respectively. Moreover, with respect to the coaxial tricopter configuration shown in Fig. 1, the expressions for total external force ($F_{ext}$) and total external moment ($\tau_{ext}$) acting in $\mathcal{F}_B$ frame, are written as:

$$F_{ext} = \begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ F_1 + F_2 + F_3 + F_4 + F_5 + F_6 \end{bmatrix}, \quad (8)$$

and

$$\tau_{ext} = \tau_{prop} + \tau_{gyro}, \tag{9}$$

where $\tau_{ext}$ is composed of $[\tau_x, \tau_y, \tau_z]^T$, and $\tau_{prop}$ and $\tau_{gyro}$ are the commonly known propeller's reaction and gyroscopic moments that are expressed as:

$$\tau_{prop} = \begin{bmatrix} \{(F_5 + F_6) - (F_1 + F_2)\} l_2 \\ (F_3 + F_4)\, l_1 - (F_1 + F_2 + F_5 + F_6)\, l_3 \\ (\tau_1 + \tau_3 + \tau_5) - (\tau_2 + \tau_4 + \tau_6) \end{bmatrix}, \tag{10}$$

$$\tau_{gyro} = J_P \begin{bmatrix} q\,\{(\Omega_2 + \Omega_4 + \Omega_6) - (\Omega_1 + \Omega_3 + \Omega_5)\} \\ p\,\{(\Omega_1 + \Omega_3 + \Omega_5) - (\Omega_2 + \Omega_4 + \Omega_6)\} \\ 0 \end{bmatrix}, \tag{11}$$

where $J_P$ represents the propeller's moment of inertia.

For a custom designed Y6 coaxial tricopter in our laboratory, the intrinsic parameters are listed in Table I. While the parameters such as mass ($m$) and arm lengths ($l_1$,$l_2$,$l_3$) are directly measured, the moment of inertia values are obtained from its SolidWorks CAD model. In addition, the thrust and drag-moment coefficients are evaluated based on a simple experimentation, details of which can be seen in [23].

For the real-time dynamic optimization framework, the nonlinear tricopter model is discretized based on the direct multiple shooting method [24], utilizing a shooting grid size of $\Delta t_s = 0.01$s. In addition, the explicit Runge-Kutta $4^{\text{th}}$ order integrator, with 2 steps per shooting interval is incorporated. Finally, the discretized nonlinear model of the coaxial tricopter UAV can be represented as:

$$\mathtt{x}_{k+1} = \mathtt{f_d}(\mathtt{x}_k, \mathtt{u}_k, \mathtt{p}), \tag{12}$$

$$\mathtt{y}_k = \mathtt{h}(\mathtt{x}_k, \mathtt{u}_k), \tag{13}$$

where the state vector $\mathtt{x} \in \mathbb{R}^{n_\mathtt{x}}$ and control vector $\mathtt{u} \in \mathbb{R}^{n_\mathtt{u}}$ are composed of:

$$\mathtt{x} = [x, y, z, u, v, w, \phi, \theta, \psi, p, q, r]^T, \tag{14}$$

$$\mathtt{u} = [\omega_1, \omega_2, \omega_3, \omega_4, \omega_5, \omega_6]^T, \tag{15}$$

and $\mathtt{y} \in \mathbb{R}^{n_\mathtt{x}}$ is the measurement vector. The term $\mathtt{p} \in \mathbb{R}^{n_\mathtt{u}}$ represents a vector containing the fault information in terms of fault parameters, i.e. $\mathtt{p} = [\alpha_i | i = 1, \cdots, 6]^T$. The state and measurement functions are denoted by $\mathtt{f_d}(\cdot,\cdot,\cdot)$:$\mathbb{R}^{n_\mathtt{x}} \times \mathbb{R}^{n_\mathtt{u}} \times \mathbb{R}^{n_\mathtt{u}} \longrightarrow \mathbb{R}^{n_\mathtt{x}}$ and $\mathtt{h}(\cdot,\cdot)$:$\mathbb{R}^{n_\mathtt{x}} \times \mathbb{R}^{n_\mathtt{u}} \longrightarrow \mathbb{R}^{n_\mathtt{x}}$, respectively.

TABLE I: Coaxial tricopter intrinsic parameters, as visualized in Fig. 1.

| Par. | Description | Value |
|---|---|---|
| $m$ | Mass of tricopter UAV | 1.412 kg |
| $l_1$ | Moment arm | 0.28 m |
| $l_2$ | Moment arm | 0.243 m |
| $l_3$ | Moment arm | 0.14 m |
| $I_{xx}$ | Moment of inertia about $\mathcal{F}_{B_x}$ | 0.014965 kg-m$^2$ |
| $I_{yy}$ | Moment of inertia about $\mathcal{F}_{B_y}$ | 0.021915 kg-m$^2$ |
| $I_{zz}$ | Moment of inertia about $\mathcal{F}_{B_z}$ | 0.034433 kg-m$^2$ |
| $I_{xz}$ | Moment of inertia about $\mathcal{F}_{B_{xz}}$ | 0.0261953 kg-m$^2$ |
| $J_P$ | Propeller's moment of inertia | $6 \times 10^{-5}$ kg-m$^2$ |
| $K_F$ | Aerodynamic force coefficient | $6.85 \times 10^{-6}$ N-s$^2$ |
| $K_\tau$ | Aerodynamic drag-moment coefficient | $3.35 \times 10^{-7}$ Nm-s$^2$ |

## III. Nonlinear Model Predictive Control

(N)MPC is an advanced, dynamic optimization-based strategy for feedback control which computes an optimum current control action by optimizing the predicted system's behavior over a finite window, often recalled as prediction horizon ($N_c$). This optimal forecast of the system's behavior is induced from an open-loop online optimization which is represented in the form of a constrained, finite horizon, optimal control problem (OCP). The open-loop OCP, when solved for the current state of the system and at the current time instant, gives an optimal sequence of open-loop control actions; the first term of which is then regarded as the optimal control action [8]. Moreover, the finite prediction window recedes forward in time, which results in a *receding horizon control* technique.

Similar to other tracking applications [25]–[28], we formulate the parametric OCP for NMPC in the form of a least square function that penalizes the deviations of predicted state ($\mathtt{x}_k$) and control ($\mathtt{u}_k$) trajectories from their specified references, over the given prediction horizon window ($t_j \leq t \leq t_{j+N_c}$). Therefore, the optimization problem in discrete time is of the form:

$$\min_{\mathtt{x}_k, \mathtt{u}_k} \quad \frac{1}{2}\Bigg\{ \sum_{k=j}^{j+N_c-1} \left( \left\|\mathtt{x}_k - \mathtt{x}_k^{\text{ref}}\right\|^2_{W_\mathtt{x}} + \left\|\mathtt{u}_k - \mathtt{u}_k^{\text{ref}}\right\|^2_{W_\mathtt{u}} \right) + \left\|\mathtt{x}_{N_c} - \mathtt{x}_{N_c}^{\text{ref}}\right\|^2_{W_{N_c}} \Bigg\} \tag{16a}$$

$$\text{s.t.} \quad \mathtt{x}_j = \hat{\mathtt{x}}_j, \tag{16b}$$

$$\mathtt{x}_{k+1} = \mathtt{f_d}(\mathtt{x}_k, \mathtt{u}_k, \mathtt{p}), \quad k = j, \cdots, j + N_c - 1, \tag{16c}$$

$$\mathtt{x}_{k,\min} \leq \mathtt{x}_k \leq \mathtt{x}_{k,\max}, \quad k = j, \cdots, j + N_c, \tag{16d}$$

$$\mathtt{u}_{k,\min} \leq \mathtt{u}_k \leq \mathtt{u}_{k,\max}, \quad k = j, \cdots, j + N_c - 1, \tag{16e}$$

where $\mathtt{x}_k \in \mathbb{R}^{n_\mathtt{x}}$ is the differential state, $\mathtt{u}_k \in \mathbb{R}^{n_\mathtt{u}}$ is the control input and $\hat{\mathtt{x}}_j \in \mathbb{R}^{n_\mathtt{x}}$ is the current state estimate; time-varying state and control references are denoted by $\mathtt{x}_k^{\text{ref}}$ and $\mathtt{u}_k^{\text{ref}}$, respectively; the terminal state reference is represented by $\mathtt{x}_{N_c}^{\text{ref}}$; $W_\mathtt{x} \in \mathbb{R}^{n_\mathtt{x} \times n_\mathtt{x}}$, $W_\mathtt{u} \in \mathbb{R}^{n_\mathtt{u} \times n_\mathtt{u}}$, and $W_{N_c} \in \mathbb{R}^{n_\mathtt{x} \times n_\mathtt{x}}$ are the corresponding weight matrices, which are assumed to be constant for simplicity. Furthermore, $\mathtt{x}_{k,\min} \leq \mathtt{x}_{k,\max} \in \mathbb{R}^{n_\mathtt{x}}$ and $\mathtt{u}_{k,\min} \leq \mathtt{u}_{k,\max} \in \mathbb{R}^{n_\mathtt{u}}$, specify the lower and upper bounds on the states and control inputs, respectively.

The last expression in (16a) represents the final cost incurred due to control variables at the end of the prediction horizon ($t_{j+N_c}$) and is generally referred as the *terminal penalty* term. This term caters for the cost incurred due to a finite horizon and is often included in the problem formulation for stability reasons [29]. Next, we present a closed-loop framework which enables the UAV to be robust against a complete actuator failure. As specified before, we utilize a cascaded NMPC structure with a position tracking high-level NMPC (H-NMPC) and an attitude tracking low-level NMPC (L-NMPC). In addition, L-NMPC is designed to be reconfigurable, such that, mere one controller is required to tackle both normal and faulty operations. The overall

closed-loop is presented in Fig. 2. It is to be noted that in this study, once the fault is induced its occurrence information is explicitly fed to the controller without any detection algorithm.

### A. High-level NMPC

The H-NMPC is designed to be responsible for tracking a given position trajectory. Utilizing the state feedback at the current instant, it computes the optimized solution in terms of total thrust and attitude angles, as can be visualized in Fig. 2. Thus, the state vector $\mathtt{x}$ and control vector $\mathtt{u}$ for H-NMPC are considered to be as follows:

$$\mathtt{x}_{\text{H-NMPC}} = [x, y, z, u, v, w]^T, \tag{17a}$$

$$\mathtt{u}_{\text{H-NMPC}} = [F_z^*, \phi^*, \theta^*, \psi^*]^T. \tag{17b}$$

Also, the following state and control reference trajectories are selected for the optimization problem of H-NMPC:

$$\mathtt{x}^{\text{ref}}_{\text{H-NMPC}} = \mathtt{x}^{\text{ref}}_{N_c,\text{H-NMPC}} = [x_r, y_r, z_r, 0, 0, 0]^T, \tag{18a}$$

$$\mathtt{u}^{\text{ref}}_{\text{H-NMPC}} = [0.1mg, 0, 0, 0]^T, \tag{18b}$$

where $m$ and $g$ are mass and gravitational constant, listed in Table I. It is to be noted that for the OCP formulation of H-NMPC, the parameterization of the nonlinear model (in translation) is also done with respect to $p$, $q$, and $r$. Hence, the three rotational rates are fed to H-NMPC along with the other states, as also illustrated in Fig. 2.

For achieving a stable response from the subsequent (low-level) controller, following constraints are introduced in H-NMPC:

$$0 \text{ (N)} \leq F_z^* \leq 0.5mg \text{ (N)}, \tag{19a}$$

$$-35 \text{ (}^\circ\text{)} \leq \phi^* \leq 35 \text{ (}^\circ\text{)}, \tag{19b}$$

$$-35 \text{ (}^\circ\text{)} \leq \theta^* \leq 35 \text{ (}^\circ\text{)}. \tag{19c}$$

In addition, its weight matrices are selected as:

$$W_{\mathtt{x},\text{H-NMPC}} = \text{diag}(30, 30, 30, 2, 2, 0.7), \tag{20a}$$

$$W_{\mathtt{u},\text{H-NMPC}} = \text{diag}(0.007, 35, 35, 40), \tag{20b}$$

$$W_{N_c,\text{H-NMPC}} = \text{diag}(50, 50, 40, 3, 3, 2), \tag{20c}$$

by the trial-and-error method such that a smooth closed-loop response of the UAV is achieved. Furthermore, the prediction window size $N_c = 30$ is selected for the H-NMPC.

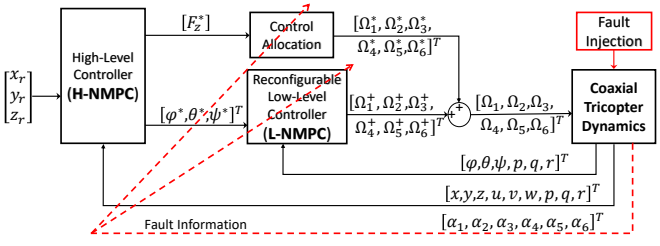



Fig. 2: Closed-loop diagram of the reconfigurable fault-tolerant NMPC framework.

**Notation:** $(\cdot)_r$ – reference; $(\cdot)^*$ – control output of H-NMPC; $(\cdot)^+$ – control output of L-NMPC; $(\cdot)$(without subscript/superscript) – feedback.

*Remark 1:* One may notice a control allocation block after the H-NMPC in Fig. 2. It computes the effective rotor velocities $[\Omega_i^* | i = 1, \cdots, 6]$, that are required to climb/descent, based upon the total thrust command $(F_z^*)$ of H-NMPC:

$$\Omega_i^* = \alpha_i \sqrt{\frac{F_z^*}{N_{\text{OR}} K_F}}, \text{ for } \quad i = 1, \cdots, 6, \tag{21}$$

where $N_{\text{OR}}$ represents the number of operational rotors having $\alpha_i = 1$. For instance, if the rotor 1 fails, the corresponding effective rotor velocities are:

$$\Omega_1^* = 0, \quad \Omega_i^* = \sqrt{\frac{F_z^*}{5K_F}}, \text{ for } \quad i = 2, \cdots, 6. \tag{22}$$

### B. Reconfigurable low-level NMPC

The state, control and parameter vectors of the low-level attitude NMPC are composed of:

$$\mathtt{x}_{\text{L-NMPC}} = [\phi, \theta, \psi, p, q, r]^T, \tag{23a}$$

$$\mathtt{u}_{\text{L-NMPC}} = \left[\omega_1^+, \omega_2^+, \omega_3^+, \omega_4^+, \omega_5^+, \omega_6^+\right]^T, \tag{23b}$$

$$\mathtt{p}_{\text{L-NMPC}} = [\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5, \alpha_6]^T, \tag{23c}$$

where the control output of H-NMPC ($\mathtt{u}_{\text{H-NMPC}}$) is given as the state reference trajectory to the L-NMPC:

$$\mathtt{x}^{\text{ref}}_{\text{L-NMPC}} = \mathtt{x}^{\text{ref}}_{N_c,\text{L-NMPC}} = [\phi^*, \theta^*, \psi^*, 0, 0, 0]^T. \tag{24}$$

In the control framework of Fig 2, one may note that the control outputs of L-NMPC $(\left[\omega_i^+ | i = 1, \cdots, 6\right])$ are directly given as the effective rotor angular velocities $(\left[\Omega_i^+ | i = 1, \cdots, 6\right])$. This is because the contributions of the fault parameters $([\alpha_i | i = 1, \cdots, 6])$ have already been taken care within the problem formulation of L-NMPC.

Subsequent to obtaining the L-NMPC's control outputs $(\Omega_i^+)$, they are added with H-NMPC's control outputs $(\Omega_i^*)$ before given to the actuators $(\Omega_i)$, as shown in Fig. 2. This is done to perform a 3-dimensional position tracking as the control output of H-NMPC only implies the thrust required for the tricopter to balance its weight (or climb/descent), while L-NMPC's control outputs facilitate the motion along $x$- and $y$-directions. Also, the following control constraints are imposed in L-NMPC such that even after full actuation command from L-NMPC, the UAV still has some room to follow H-NMPC's command without saturating the rotors:

$$0 \leq \omega_1^+ \leq 900, \quad 0 \leq \omega_2^+ \leq 900, \quad 0 \leq \omega_3^+ \leq 900, \tag{25a}$$

$$0 \leq \omega_4^+ \leq 900, \quad 0 \leq \omega_5^+ \leq 900, \quad 0 \leq \omega_6^+ \leq 900, \tag{25b}$$

where $\omega_i^+$ are in rad/s. Furthermore, the prediction window size $N_c = 30$ is selected to facilitate the real-time applicability of the proposed closed-loop. Next, we illustrate the design of L-NMPC for the two working scenarios, i.e., normal operation and faulty operation.

*1) Normal Operation:* When all the rotors are functioning $(N_{\text{OR}} = 6)$, L-NMPC equally distributes the control load amongst all rotors. Therefore, the following control reference

trajectory along with the weight matrices are utilized for the optimization problem of L-NMPC:

$$\mathrm{u}^{\mathrm{ref}}_{\text{L-NMPC}} = [486, 486, 486, 486, 486, 486]^T, \tag{26a}$$

$$W_{\mathrm{x,L\text{-}NMPC}} = \mathrm{diag}(30, 25, 30, 0.1, 2, 0.1), \tag{26b}$$

$$W_{\mathrm{u,L\text{-}NMPC}} = \mathrm{diag}(10^{-5}, 10^{-5}, 10^{-5}, 10^{-5}, 10^{-5}, 10^{-5}), \tag{26c}$$

$$W_{N_c,\text{L-NMPC}} = \mathrm{diag}(120, 110, 80, 0.1, 0.1, 0.1), \tag{26d}$$

where the above weights are obtained by the trial-and-error method.

*2) Faulty Operation:* When a rotor fault is induced, the information regarding its location, in terms of vector p, is given to L-NMPC which subsequently reconfigures itself. In this reconfiguration phase, the L-NMPC solves the optimization problem of (16) such that the optimized solution is computed by considering only the operational rotors (with $\alpha_i = 1$). For the two sequential rotor failure cases treated in this study, the following modified input reference trajectories and weight matrices are utilized within the reconfiguration phase:

- Rotor 3 failure:

$$\mathrm{u}^{\mathrm{ref}}_{\text{L-NMPC}} = [595, 344, 0, 687, 595, 344]^T, \tag{27a}$$

$$W_{\mathrm{x,L\text{-}NMPC}} = \mathrm{diag}(100, 105, 25, 0.1, 1.5, 0.6), \tag{27b}$$

$$W_{\mathrm{u,L\text{-}NMPC}} = \mathrm{diag}(3\times10^{-5}, 3\times10^{-5}, 10^{+4}, 3\times10^{-5}, 3\times10^{-5}, 3\times10^{-5}), \tag{27c}$$

- Rotor 6 failure:

$$\mathrm{u}^{\mathrm{ref}}_{\text{L-NMPC}} = [344, 595, 344, 595, 687, 0]^T, \tag{28a}$$

$$W_{\mathrm{x,L\text{-}NMPC}} = \mathrm{diag}(150, 125, 20, 0.1, 1.5, 0.5), \tag{28b}$$

$$W_{\mathrm{u,L\text{-}NMPC}} = \mathrm{diag}(3\times10^{-5}, 3\times10^{-5}, 3\times10^{-5}, 3\times10^{-5}, 3\times10^{-5}, 10^{+4}), \tag{28c}$$

while the terminal penalty weights ($W_{N_c,\text{L-NMPC}}$) are kept the same as specified in (26d). The reason to select these two rotor failure cases is their location in the coaxial tricopter frame, which helps in the generalization of the obtained results.

## IV. Numerical Results

In this section, we present the simulation results for the proposed cascaded NMPC framework which is employed for the FTC application of a coaxial tricopter UAV. The optimization problem of NMPC in (16) is solved using the direct multiple shooting method and real-time iteration (RTI) approach incorporated in ACADO toolkit. It is an open-source C++ based software environment that creates self-contained C codes, utilizing its code generation package [30]. These C codes can be exported and executed on any computational platform including low-cost embedded hardware. In our application, we utilize these C codes in order to create MATLAB/Simulink based S-functions. Moreover, all the simulations are performed in MATLAB/Simulink installed on a hardware consisting of 3.4-GHz Intel Core i7 processor and 8 GB RAM.

For initializing the simulation, the UAV is considered to be on the ground with zero state and control vectors, and with the parameters given in Table I. A fixed sampling time of $\Delta t = 0.01$s is maintained for the entire simulation. In the subsequent part, we provide the position tracking results for an 8-shaped reference trajectory. The tricopter first climbs to 1.5m height and hovers there for 6s. Then, the position reference commands ($[x_r, y_r, z_r]$) are given to the H-NMPC till the end of the simulation time, i.e., 95s. The normal operation is considered for the first 35s and then, the fault is induced in rotor 6 ($\alpha_6 = 0$) for 20s. Thereafter, the normal operation resumes for another 10s and subsequently, rotor 3 fails ($\alpha_3 = 0$) for 20s, followed by the normal operation until the end. The instants of fault and normal operations are represented by dotted black and cyan vertical lines, respectively, in all the figures. Since a fault-detection algorithm in not explicitly utilized in this work, a delay time of 0.5s for fault-detection is explicitly considered in the simulation results.

### *8-Shaped Reference*

An 8-shaped trajectory, 4m wide along $x$-direction and 2m wide along $y$-direction, is generated for the UAV after it climbs to 1.5m in $z$-direction. For the selected weights of H-NMPC and L-NMPC, the tracking performance of the closed-loop framework in the presence of intermittent faults can be seen in Fig. 3. Although the UAV deviates slightly

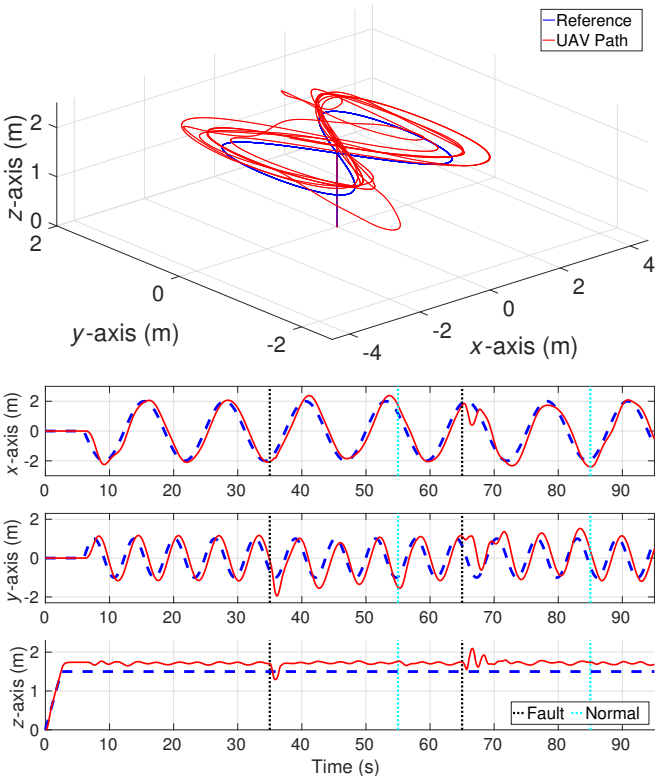
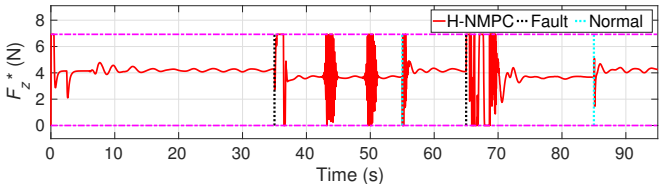



Fig. 3: 3-dimensional position tracking performance of the UAV with two sequential rotor failure cases. Even with a complete rotor failure, UAV manages to track the trajectory.

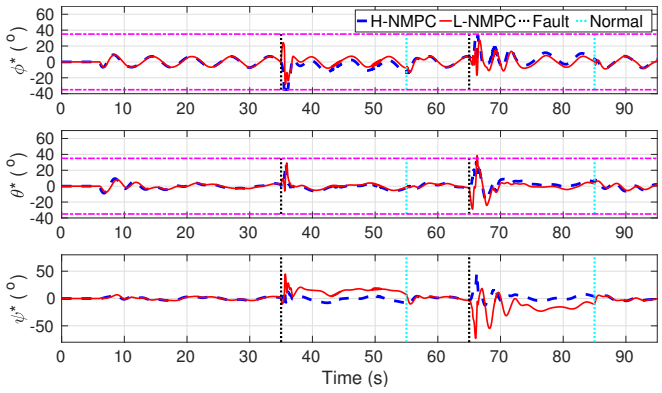
from the commanded path at the very next instant of fault occurrence, it is able to recover within few seconds. In other words, the UAV does not leave its course even after experiencing sequential rotor failures but continues to track the commanded trajectory. This validates the main goal of achieving FTC of the coaxial tricopter UAV. Additionally, its overall $x$-, $y$-, and $z$-direction tracking root-mean-squared error (RMSE) values for the 8-shaped reference are $0.440$m, $0.544$m, and $0.223$m, respectively.

The thrust and attitude angles commanded by H-NMPC are shown in Figs. 4a and 4b, respectively, where the dotted magenta lines represent the constraints that are specified in (19). As evident in Fig. 4, none of the constraints are ever violated by H-NMPC, even with the sudden jumps that arise at the instants of actuator failures, which eventually supports the real-time applicability of the proposed closed-loop framework. Thereafter, the control inputs commanded by L-NMPC are shown in Fig. 5, where the dotted magenta lines again represent the constraints that are specified in (25). It is illustrated from Fig. 5 that in the event of an actuator failure, the L-NMPC successfully reconfigures itself such that with the remaining actuators UAV still can continue to follow the desired trajectory. The RMSE values in tracking $\phi$, $\theta$ and $\psi$ attitude angles by L-NMPC for the 8-shaped trajectory are $4.707°$, $3.599°$ and $13.409°$, respectively. Moreover, the final actuator inputs given to the tricopter rotors are depicted in Fig. 6.

In order to analyze the effect of fault-detection delay time on overall tracking accuracy, the mean Euclidean error, and maximum altitude gain/loss ($\Delta z$) due to each failure, are listed in Table II. As anticipated, the tracking accuracy decreases with the increasing delay time. Moreover, in this




(a) Thrust output




(b) Attitude angles output

Fig. 4: Controller outputs for H-NMPC with the actual response of the UAV.

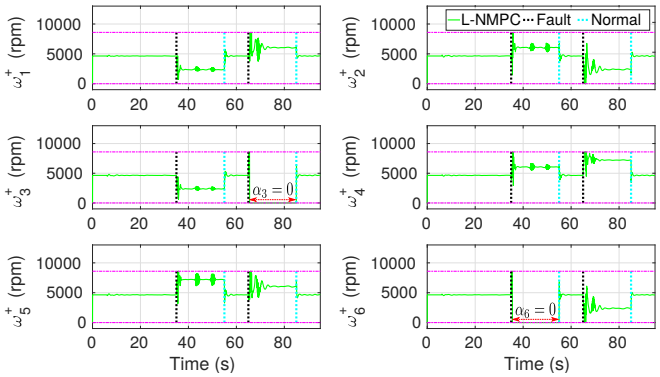



Fig. 5: Controller outputs for L-NMPC in terms of rotor speeds. None of the constraints are violated even during fault instances.

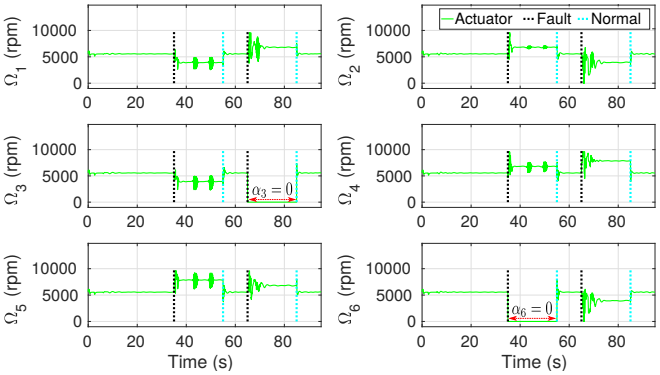



Fig. 6: Actuator inputs given to the tricopter rotors after a summation between H-NMPC ($\Omega_i^*$) and L-NMPC ($\Omega_i^+$) effective rotor velocities.

application, the reconfigurable L-NMPC is designed to tackle the delay time upto $0.5$s, which is sufficiently large for fast UAV applications.

*Remark 2:* During the controller tuning process, it is observed that the failure of back rotor is most critical for the considered coaxial tricopter configuration. This is in accordance with our intuition as the pitch control gets severely affected due to the unavailability of sufficient torque which is required to balance the torque produced by the four front rotors.

The average execution times of H-NMPC and L-NMPC are $0.52$ms and $1.58$ms, respectively. This implies that the combined average execution time of the closed-loop is around $2.1$ms, which is substantially less than the adopted

TABLE II: Performance variation (unit: m) with different fault-detection delay times.

| | **0.1s** | **0.2s** | **0.3s** | **0.4s** | **0.5s** |
|---|---|---|---|---|---|
| **Mean Euclidean error** | 0.649 | 0.650 | 0.651 | 0.659 | 0.674 |
| **Max. $\Delta z$ change for $\alpha_6 = 0$** | 0.046 | 0.131 | 0.251 | 0.359 | 0.408 |
| **Max. $\Delta z$ change for $\alpha_3 = 0$** | 0.015 | 0.033 | 0.145 | 0.340 | 0.349 |

sampling time of 10ms, and thus, justify the real-time feasibility of the cascaded NMPC framework for FTC applications.

## V. Conclusions

In this paper, a cascaded NMPC framework has been proposed for controlling a coaxial tricopter UAV in the event of complete loss of one rotor. The enticing prospect of this closed-loop framework is the reconfigurable L-NMPC that eliminates the need to utilize multiple controllers for normal and faulty operations, and thus, significantly reduces the overall control design complexity. In the reconfiguration phase, L-NMPC computes the optimized solution by considering only the operational rotors. The fault-tolerant capability of the closed-loop has been illustrated for tracking an 8-shaped trajectory, and a satisfactory performance has been obtained while accommodating fault-detection delay times upto 0.5s. What is more, owing to ACADO toolkit, a combined average execution time of 2.1 milliseconds is achieved that demonstrates the real-time-feasibility of the proposed closed-loop framework for fault-tolerant aerospace applications.

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